Note 8.5

Suppose three sides are given in vector form, prove

(i) either sum of the vectors is $\vec{0}$ or sum of any two vectors is equal to the third vector, to form a triangle.

(ii) dot product between any two vectors is 0 to ensure one angle is $\frac{\pi}{2}$.

EXERCISE 8.3

(1) Find $\vec{a}.\vec{b}$ when

(i) $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$ and $\vec{b}=3\hat{i}-4\hat{j}-2\hat{k}$ (ii) $\vec{a}=2\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=6\hat{i}-3\hat{j}+2\hat{k}$.

(2) Find the value λ for which the vectors $\vec{a}$ and $\vec{b}$ are perpendicular, where

(i) $\vec{a}=2\hat{i}+\lambda\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+3\hat{k}$ (ii) $\vec{a}=2\hat{i}+4\hat{j}-\hat{k}$ and $\vec{b}=3\hat{i}-2\hat{j}+\lambda\hat{k}$.

(3) If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}|=10, |\vec{b}|=15$ and $\vec{a}\cdot\vec{b}=75\sqrt{2}$, find the angle between $\vec{a}$ and $\vec{b}$.

(4) Find the angle between the vectors

(i) $2\hat{i}+3\hat{j}-6\hat{k}$ and $6\hat{i}-3\hat{j}+2\hat{k}$ (ii) $\hat{i}-\hat{j}$ and $\hat{j}-\hat{k}$.

(5) If $\vec{a},\vec{b},\vec{c}$ are three vectors such that $\vec{a}+2\vec{b}+\vec{c}=\vec{0}$ and $|\vec{a}|=3, |\vec{b}|=4, |\vec{c}|=7$, find the angle between $\vec{a}$ and $\vec{b}$.

(6) Show that the vectors $\vec{a}=2\hat{i}+3\hat{j}+6\hat{k}$, $\vec{b}=6\hat{i}+2\hat{j}-3\hat{k}$, and $\vec{c}=3\hat{i}-6\hat{j}+2\hat{k}$ are mutually orthogonal.

(7) Show that the vectors $-\hat{i}-2\hat{j}-6\hat{k}, 2\hat{i}-\hat{j}+\hat{k}$, and $-\hat{i}+3\hat{j}+5\hat{k}$ form a right angled triangle.

(8) If $|\vec{a}|=5$, $|\vec{b}|=6$, $|\vec{c}|=7$ and $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, find $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$.

(9) Show that the points (2, – 1, 3), (4, 3, 1) and (3, 1, 2) are collinear.

(10) If $\vec{a},\vec{b}$ are unit vectors and θ is the angle between them, show that

(i) $\sin\frac{\theta}{2}=\frac{1}{2}|\vec{a}-\vec{b}|$ (ii) $\cos\frac{\theta}{2}=\frac{1}{2}|\vec{a}+\vec{b}|$ (iii) $\tan\frac{\theta}{2}=\frac{|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|}$.

(11) Let $\vec{a},\vec{b},\vec{c}$ be three vectors such that $|\vec{a}|=3$, $|\vec{b}|=4$, $|\vec{c}|=5$ and each one of them being perpendicular to the sum of the other two, find $|\vec{a}+\vec{b}+\vec{c}|$.

(12) Find the projection of the vector $\hat{i}+3\hat{j}+7\hat{k}$ on the vector $2\hat{i}+6\hat{j}+3\hat{k}$.

(13) Find λ, when the projection of $\vec{a}=\lambda\hat{i}+\hat{j}+4\hat{k}$ on $\vec{b}=2\hat{i}+6\hat{j}+3\hat{k}$ is 4 units.

(14) Three vectors $\vec{a},\vec{b}$ and $\vec{c}$ are such that $|\vec{a}|=2$, $|\vec{b}|=3, |\vec{c}|=4$, and $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. Find $4\vec{a}\cdot\vec{b}+3\vec{b}\cdot\vec{c}+3\vec{c}\cdot\vec{a}$.

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